ALPACA AI TRADING AGENTS HACKATHON

Contour — the measurement picks the structure

`vrp_ratio < 1.30` → **NO_TRADE** implied is not rich enough to sell

`skew_z ≥ +0.8` → **PUT_CREDIT_SPREAD** puts rich — sell puts, not cheap calls

`skew_z ≤ −0.8` → **CALL_CREDIT_SPREAD** calls rich — sell calls, not cheap puts

`otherwise` → **IRON_CONDOR** both sides fair — sell both

AI LOGIC — THE LEASH

Every wired model output can only make the agent trade **less**. Not a promise — a property of the import graph.

THE MODEL MAY	THE MODEL MAY NEVER
• Name event windows to stand down in	• Choose a strike
• Veto a proposed structure	• **Size** or price a position
• Stand the whole book down	• Reverse or widen anything

`execute.py never imports mind.py`

GLM-5 on Amazon Bedrock · picked by bake-off

ALPACA INFRASTRUCTURE

- **Orders go through the CLI, and that is a finding.** The MCP server cannot place multi-leg orders — the `legs` array arrives as a JSON string and fails validation. `alpaca-mcp-server#97`, open since July. The CLI places the identical 4-leg order correctly — and we sent the fix upstream as `#118`.
- Reads are `alpaca-py`: chain snapshots carry Greeks but no `open_interest`, so they are merged with Trading API contract objects.
- Entries are a **three-rung limit ladder** from mid toward the bid, never a market order. `reconcile()` reads actual per-leg fills: paper issues random partials, and trusting the request puts condor legs out of ratio.
- The journal is an **append-only SHA-256 hash chain**. The dashboard recomputes it in your browser and prints the same verdict the CLI does.
- Autonomy is GitHub Actions: pre-open planning, a cycle every 15 minutes, a scheduled Thursday flatten. **No `pull_request` trigger** on the credentialed workflow.
- **No resting stop exists on a multi-leg position** — so the options book polls. A single equity leg *can* rest one, and the directional sleeve parks its stop GTC at the broker: the only exit that works overnight.

Twelve risk gates, plus seven for the sleeve — every reason journaled, pass or fail

296 tests

The volatility premium is real, and it is **not uniform**. That dispersion is the entire opportunity.

SPY	QQQ	IWM
1.55	1.23	1.17
paid to sell	not paid enough	not paid enough

- Short premium is the crowded retail options trade — and the crowd sells **one structure regardless of the surface**.
- **Friction decides the universe, not opinion.** Single-name weeklies cost `$40–80` round trip against a `$30–42` modelled edge — it loses to costs before it starts. Three ETFs cost `$8–20`.
- Alpaca serves **no earnings-date endpoint on any plan**, so a single-name design hangs its most important gate on data that does not exist.

implied / 10-day realized, measured by the live agent

floor: vrp_ratio 1.30

COMPETITIVE ANALYSIS

SUBMISSION	ITS OWN DESCRIPTION
Horizon Blackline	LLM proposes, deterministic risk gates authorize, hash-chained and auditable
VRP Engine	Harvests the variance risk premium with defined-risk spreads and risk gates
AEGIS-Q	Bounded AI selects a pre-validated bullish or bearish spread — or abstains
EdgeStack	Journals every trade and every refusal
Contour	**Chooses which of four structures to sell, from measured 25-delta skew.** And is runnable without our credentials.

“LLM proposes, gates dispose, everything journaled” is table stakes now

`python -m contour --replay`

PERFORMANCE, ATTRIBUTED

The criterion asks for the P&L of **the submitted agent**. This account holds two traders, and the broker records which is which. 2026-09-04 04:01 UTC

PLACED BY	P&L	OF START NAV
The agent — every id it chose	**+$147.55**	**+0.15%**
The operator — three discretionary tail trades	−$642.00	−0.64%
Account total	−$494.45	−0.49%

Not our bookkeeping — a field the broker stamps. Entries are prefixed `contour-` by `loop.py`; exits are named after the entry they close. Nothing else in the account is. A symbol touched by both is charged to the operator — the published number is the pessimistic one.

The systematic book is flat in a week QQQ fell 1.27% in a session

`python ops/attribution.py --offline`

REVENUE MODEL

The audit layer is the product

The gate engine and the hash-chained decision record, licensed to brokers and RIAs who have to **defend** an automated decision after the fact. That is the durable asset here — not the alpha, which decays.

Own capital

P&L is the revenue and no registration question arises. The honest ceiling on defined-risk premium selling is the credit, so this scales with capital, not with claims — and where we stepped outside that ceiling, the write-up names the trade and its negative expectancy.

Not signal subscriptions. Selling trade recommendations is investment advice and needs registration — saying so out loud is a credibility position, not a limitation.

median week: under +1%

SHIPPED

- 3 ETFs, one locked expiry, 15-minute cycle
- **A $30k long-QQQ sleeve** beside the options book — variance, not edge — funded *out of* the same −4% floor, not beside it
- **A $4.4k long-call tail**, added *and closed* 2026-09-01 — the one negative-EV trade here, labelled as such. Sold for $3,113, realising **−$1,320**: the entire drawdown. Closed once a 387-cycle backtest found no edge in the book worth paying ~15% over fair value to lever, and the write-up says that too
- **A $1.1k TQQQ call tail**, placed on instruction after the evidence against it was recorded — a gap-down bounce tests as noise (t = +0.42). It breaks no gate, but it used the last room in front of the floor, so the entry ramp is *closed to zero* for the rest of the contest. The floor was kept by stopping, not by paying it

NEXT

- **Trend-aware structure selection.** The map breaks ties toward the condor — which sells calls into a confirmed uptrend. Three trend systems say that is the wrong default; skew alone should not decide
- Expiry laddering and rolls
- Skew priors **learned** per underlying instead of hard-coded
- **More backtest history.** The harness is built and has run — 387 cycles against real historical option prices, importing the agent's own selection and gate code rather than a reimplementation. Alpaca's option history starts 2024-01-18, so 2.5 years is the entire available sample, and t = +0.37 is what that sample supports
- Portfolio vega and gamma caps, not per- loss alone